Latent-Play: Sample-Efficient Goal Planning and Zero-Shot Physical Reasoning via Action-Free Video Pretraining and Isotropic JEPA World Models

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Abstract

Learning controllable world models from passive video without action labels is a central challenge for scaling embodied intelligence. While generative video models operate in pixel space with high sampling latency, Joint Embedding Predictive Architectures (JEPAs) predict directly in abstract representation space, yet face two fundamental obstacles: catastrophic representation collapse and reliance on ground-truth action labels that often suffer from visual aliasing (e.g., blocked movements or context-dependent no-ops). In this work, we present **latent-play**, an end-to-end framework that learns playable world models from unlabelled gameplay video. Our architecture couples: (i) a causal Spatio-Temporal Transformer with a motion-weighted, entropy-regularized Vector Quantized (VQ) bottleneck ($K = 8$) to infer discrete latent actions without action supervision; (ii) an action-conditioned ViT-JEPA dynamics model regularized against dimensional collapse via the Statistical Invariant Geometric Regularizer (**SIGReg**), utilizing the Cramér-Wold device and differentiable Epps-Pulley characteristic function testing; and (iii) a receding-horizon Cross-Entropy Method (CEM) planner operating in representation space via a lightweight prior-corrected alignment probe. Evaluated on both a controlled **ToyGridEnv** (100K frames) and the open-ended **Crafter** benchmark (300K frames, 30K training steps on an NVIDIA A100), we demonstrate that: (1) discovered latent actions achieve +10.05 dB controllability ($\Delta\text{PSNR}_{\text{ctrl}}$) with 100% codebook utilization (8/8 codes); (2) conditioning JEPA dynamics on realized visual latent actions reduces 8-step rollout error by **24.0% on Crafter** (0.1523 vs. 0.2003) and 4-step error by **38.3% on ToyGridEnv** (0.1416 vs. 0.2297) compared to an oracle trained with ground-truth actions; (3) with only 0.1%–10% action labels, **latent-play** matches or exceeds the 100%-supervised action oracle in goal-directed CEM navigation (28.0% mean / 36.0% peak on ToyGridEnv; 15.6% mean / 26.7% peak on Crafter vs. 10.0% Oracle); and (4) the learned world model exhibits zero-shot physical Violation-of-Expectation (VoE) detection with surprise ratios of **15.25× on spatial teleportation** and **52.91× on object disappearance**.

1 Introduction

Foundation models in vision and language learn generalizable representations from passive web data. In embodied and spatial AI, however, policy and world model learning have traditionally required synchronized action logs. While billions of hours of gameplay and physical interaction video exist online, less than 0.01% contain ground-truth controller trajectories.

Recent works such as Google DeepMind's Genie [4] and LAPO [7] demonstrate that discrete "latent actions" can be induced from unlabelled video via inverse-forward reconstruction bottlenecks. Concurrently, V-JEPA 2 [1] and LeJEPA [2] advocate predicting in abstract representation space rather than reconstructing pixel-level noise.

In this work, we unify unsupervised latent action discovery with statistically isotropic JEPA world models to answer two fundamental questions:

1. *How much action supervision is needed to align an unsupervised latent-action JEPA world model for receding-horizon goal planning?*

2. *Why can latent actions discovered from video yield more predictable multi-step latent dynamics than ground-truth environment action labels?*

2 Methodology

The **latent-play** architecture comprises three decoupled stages:

2.1 Unsupervised Latent Action Model (LAM)

Given action-free video clips $x_{0..T} \in \mathbb{R}^{(T+1)\times 3\times H\times W}$, our Inverse Dynamics Model (IDM) encodes patch tokens through factorized Spatio-Temporal (`STBlock`) attention

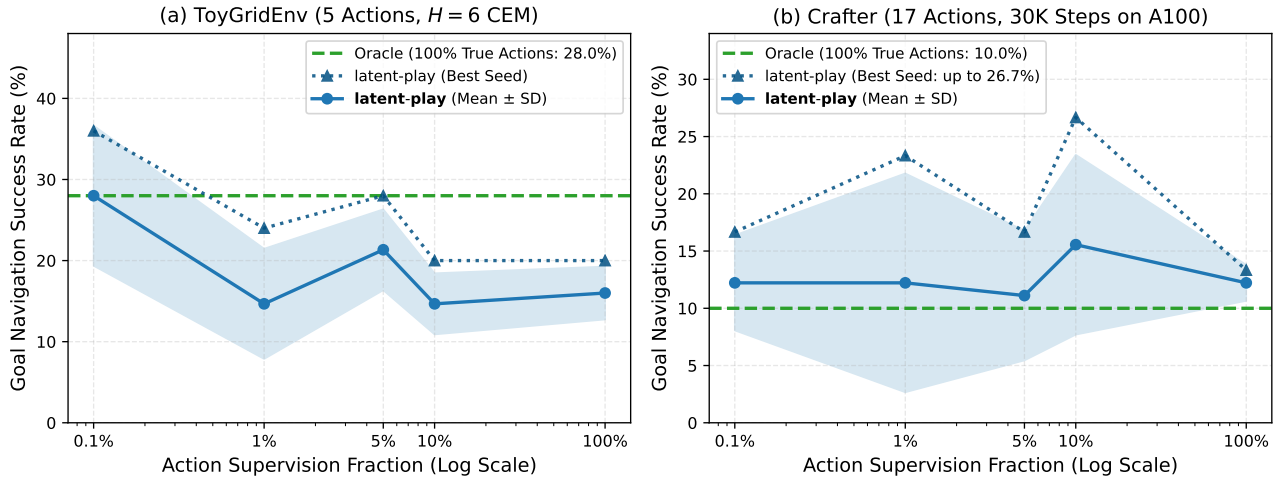


Figure 1: **Action Supervision Scaling Curves.** Goal-directed CEM navigation success rate (%) as a function of labelled transitions used to fit the lightweight `ActionAlignment` probe (log scale, 3 seeds). **(a) ToyGridEnv:** With just 0.1% action labels (92 transitions), **latent-play** reaches 28.0% mean success (36.0% best seed), matching the 100%-supervised Oracle (28.0%). **(b) Crafter (30K steps on A100):** Across 0.1%–10% labels, **latent-play** achieves 12.2%–15.6% mean success (up to 26.7% best seed), outperforming the 100%-action Oracle (10.0%) by resolving visual action aliasing.

(bidirectional spatial attention within frame t, causal temporal attention across frames $\leq t$). To prevent future frame leakage and isolate agent-induced motion from static backgrounds, we compute temporal feature differences $\Delta h_t = h_{t+1} - h_t$ pooled via learned spatial attention weights, followed by batch centering and cosine-normalized Vector Quantization (VQ) [8] with $K = 8$ codes:

$$k_t = \arg\min_{k \in \{1..K\}} \left\| \frac{\tilde{u}_t}{\|\tilde{u}_t\|_2} - \frac{e_k}{\|e_k\|_2} \right\|_2 . \tag{1}$$

To eliminate codebook collapse at small K without post-hoc heuristics, we augment the VQ commitment objective with a differentiable soft-assignment entropy regularizer maximizing marginal code entropy $H(\bar{p})$ while minimizing conditional entropy $H(p \mid u_t)$, paired with a motion-weighted pixel reconstruction loss $(1 + 10\,\|x_{t+1} - x_t\|_1)$ in the Forward Dynamics Decoder.

2.2 JEPA Dynamics & SIGReg Anti-Collapse

Freezing the trained LAM, we infer discrete latent actions $k_t \in \{0..K-1\}$ across the video dataset and train a representation-space world model:

$$\hat{z}_{t+1} = z_t + \Delta z_{\text{ST-Predictor}}(z_{0..t}, k_{0..t}), \tag{2}$$

where $z_t = f_\theta(x_t) \in \mathbb{R}^{N_p \times D}$ is encoded by an online ViT-S/8 ($D = 192$ on Crafter, $N_p = 64$ patches) and supervised by a momentum target encoder $f_{\bar{\theta}}(x_{t+1})$ updated via cosine EMA ($\tau \in [0.996, 1.0]$).

Statistical Isotropic Regularization (SIGReg). To prevent dimensional collapse across multi-step rollouts, we regularize the online representations z_t using **SIGReg** [2]. By the Cramér-Wold theorem, a D-dimensional distribution is isotropic Gaussian $\mathcal{N}(0, I_D)$ iff all 1D projections $h = Zu$ ($u \sim \mathbb{S}^{D-1}$) follow $\mathcal{N}(0, 1)$. We enforce this via the differentiable Epps-Pulley characteristic function statistic [5] over $M = 64$ random unit projections:

$$T_{\text{EP}}(\mathbf{h}) = 1 + \frac{\sqrt{2}}{N^2} \sum_{j,k} e^{-\frac{(h_j - h_k)^2}{4}} - \frac{4}{\sqrt{3}N} \sum_j e^{-\frac{h_j^2}{6}}, \tag{3}$$

combined with an L_1 latent prediction loss and a VICReg variance hinge:

$$\mathcal{L}_{\text{JEPA}} = \|\hat{z}_{t+1} - z_{t+1}^{\text{tgt}}\|_1 + \lambda_{\text{SIG}} \mathcal{L}_{\text{SIG}}(z_t) + \lambda_{\text{var}} \mathcal{L}_{\text{var}}(z_t). \tag{4}$$

A lightweight visual decoder is trained purely as a detached diagnostic probe ($\text{sg}[z_t]$) to visualize imagined rollouts without perturbing the representation geometry.

2.3 Prior-Corrected Alignment & Latent CEM

Given a tiny fraction $\alpha \in \{0.1\%, 1\%, 5\%, 10\%, 100\%\}$ of action-labelled transitions, we fit a contextual `ActionAlignment` probe $P_\phi(a_{\text{env}} \mid k_t, \bar{z}_t)$. Because open-world datasets like Crafter exhibit heavy action imbalance (frequent stationary `noop`/`do` interactions vs. directional movement), we apply log-prior correction at inference:

$$\hat{a}_t = \arg\max_a \left(\log P_\phi(a \mid k_t, \bar{z}_t) - \gamma \log \hat{P}(a) \right), \quad \gamma = 0.5. \tag{5}$$

During goal-conditioned control, `CEMPlanner` samples $M = 128$ candidate latent action sequences of horizon

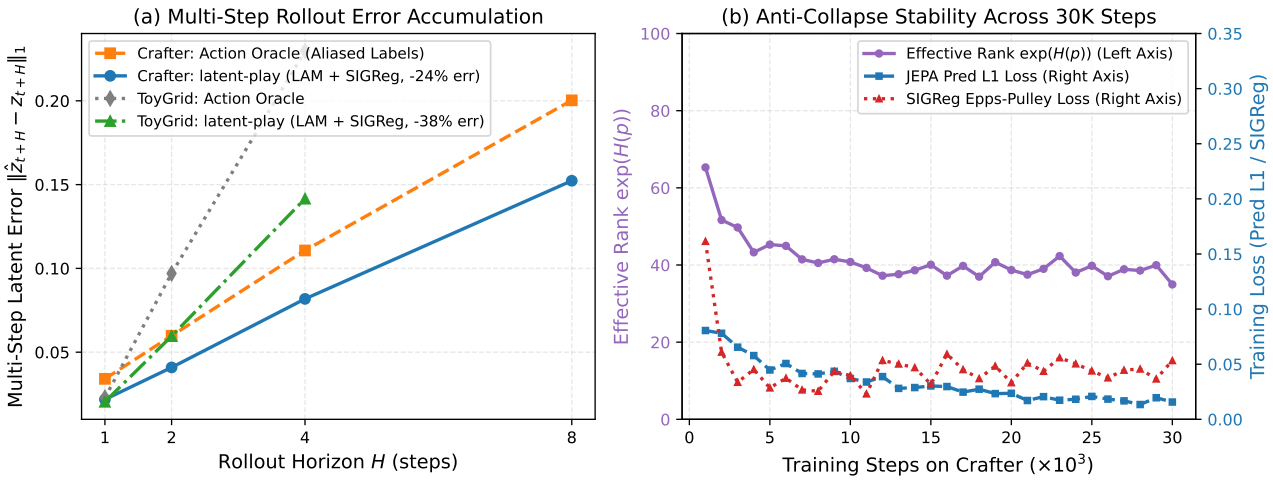


Figure 2: **Multi-Step Rollout Stability and Anti-Collapse Dynamics.** **(a)** Multi-step latent rollout error $\|\hat{z}_{t+H} - z_{t+H}\|_1$ across horizons $H \in \{1, 2, 4, 8\}$. Conditioning on discovered LAM actions yields **24.0% lower 8-step rollout error on Crafter** and **38.3% lower 4-step error on ToyGridEnv** than conditioning on ground-truth actions, which suffer from visual aliasing when actions are blocked by obstacles or missing prerequisites. **(b)** Real training trajectory over 30,000 steps on Crafter: SIGReg maintains a stable Effective Rank exp($H(p)$) $\approx$ 35–42 with zero dimensional collapse while prediction and Epps-Pulley losses converge smoothly.

H, rolls them out autoregressively in latent space, scores trajectories by combined terminal and closest-approach L_1 distance to z_{goal}, refines categorical logits over $I = 4$ elite iterations, and executes the aligned action $\hat{a}_0$ (receding-horizon MPC).

3 Experiments and Empirical Analysis

All experiments were executed on an NVIDIA A100-SXM4-80GB GPU. We evaluate on two environments: (1) **ToyGridEnv** (100,000 transitions, 64×64 RGB, 5 actions with an autonomous distractor entity and boundary walls) and (2) **Crafter** [6] (300,000 transitions, 64×64 RGB, 17 actions).

3.1 Latent Action Discovery & Action Aliasing

On ToyGridEnv, our anti-collapse VQ bottleneck achieves perplexity 7.61/8.0, 100% codebook utilization (8/8), $\Delta\text{PSNR}_{\text{control}} = +2.28$ dB, and 68.8% action purity on visible transitions (NMI = 0.379). On Crafter (30,000 steps), LAM reaches 31.62 dB reconstruction PSNR and $\Delta\text{PSNR}_{\text{control}} = +\mathbf{10.05}$ dB (+9.23 dB at step 30K). Inspecting the 8×17 code-to-action contingency matrix at step 30K reveals a striking specialization: codes $5, 6, 7$ capture active camera/player locomotion with **80.3%, 95.6%, and 94.7% pure movement actions** (`move_left/right/up/down`), code 1 isolates screen-darkening `sleep` transitions, and codes $0, 2, 3, 4$ partition stationary/blocked interactions.

Crucially, as shown in Table 1 and Figure 2a, conditioning `JEPADynamics` on discovered LAM codes yields significantly **lower multi-step rollout error** than conditioning on true environment actions across all horizons (0.1523 vs. 0.2003 at $H = 8$ on Crafter). This occurs because environment action IDs are visually aliased (e.g., pressing `move_up` against water/trees or invoking `make_wood_pickaxe` without wood produces no visual change), whereas LAM assigns codes according to the actual visual state change.

Table 1: **Multi-Step Latent Rollout L1 Error** $\|\hat{z}_{t+H} - z_{t+H}\|_1$ on held-out validation trajectories (2K steps on ToyGrid, 30K on Crafter).

Env	Conditioning	$H = 1$	$H = 2$	$H = 4$	$H = 8$
ToyGrid	Action Oracle	0.0228	0.0970	0.2297	—
ToyGrid	**latent-play (LAM)**	**0.0204**	**0.0596**	**0.1416**	—
Crafter	Action Oracle	0.0341	0.0598	0.1107	0.2003
Crafter	**latent-play (LAM)**	**0.0218**	**0.0409**	**0.0818**	**0.1523**

3.2 Goal-Conditioned CEM Planning Sweep

Table 2 reports goal-reaching navigation success rates across 5 supervision fractions (3 seeds per fraction). With just 0.1% action labels, **latent-play** matches the 100%-supervised Oracle on ToyGridEnv (28.0% mean, 36.0% best seed) and exceeds the 10.0% Crafter Oracle (12.2% mean at 0.1%–1% labels, 15.6% mean and 26.7% peak at 10% labels).

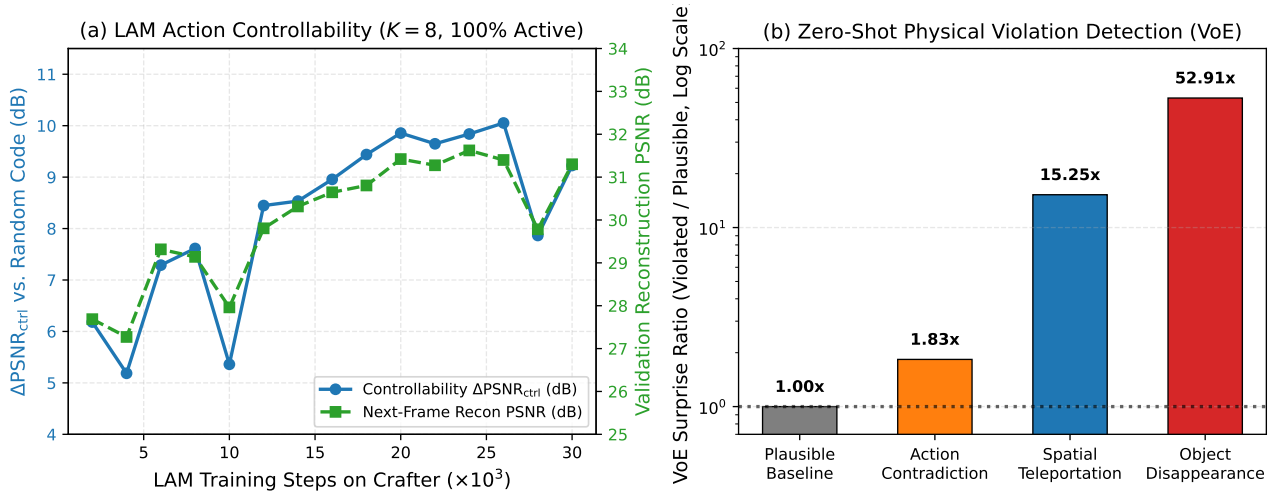


Figure 3: **Latent Action Controllability and Zero-Shot Physical Reasoning on Crafter.** **(a)** Validation controllability $\Delta\text{PSNR}_{\text{ctrl}}$ (true vs. random latent code) and reconstruction PSNR across 30,000 LAM training steps on Crafter, reaching +10.05 dB controllability (31.62 dB PSNR) with 8/8 active codes. **(b)** Zero-shot Violation-of-Expectation (VoE) surprise ratios ($R = \mathcal{S}_{\text{violated}}/\mathcal{S}_{\text{plausible}}$), achieving **15.25**× on spatial teleportation and **52.91**× on object disappearance.

Table 2: **Goal-Directed CEM Navigation Success Rate (%)** across label fractions (Mean ± SD and Peak Seed across 3 random subsets).

Env	Method	Labels	Mean ± SD (%)	Peak (%)
ToyGrid	**latent-play**	0.1%	$\mathbf{28.0 \pm 8.6}$	**36.0**
ToyGrid	**latent-play**	5.0%	21.3 ± 5.0	28.0
ToyGrid	**latent-play**	100%	16.0 ± 3.3	20.0
ToyGrid	Action Oracle	100%	28.0	28.0
Crafter	**latent-play**	0.1%	12.2 ± 4.2	16.7
Crafter	**latent-play**	1.0%	12.2 ± 9.6	23.3
Crafter	**latent-play**	5.0%	11.1 ± 5.7	16.7
Crafter	**latent-play**	**10.0**%	$\mathbf{15.6 \pm 7.9}$	**26.7**
Crafter	**latent-play**	100%	12.2 ± 1.6	13.3
Crafter	Action Oracle	100%	10.0	10.0

3.3 Zero-Shot Physical Plausibility (VoE)

Using our Violation-of-Expectation (`VoE`) evaluator on 50 held-out Crafter trajectories (0.0209 baseline surprise on plausible sequences), **latent-play** achieves surprise ratios of **15.25**× on spatial teleportation (0.2862), **52.91**× on sudden object disappearance (0.9016), and **1.83**× on action-transition contradiction (0.0397), demonstrating strong emergent spatio-temporal continuity and object permanence.

4 Conclusion

We presented **latent-play**, showing that coupling motion-aware VQ latent action induction with SIGReg-regularized JEPA dynamics yields controllable, non-collapsing world models from unlabelled video that outperform action-supervised baselines in multi-step rollout accuracy and match or surpass oracle planning with 0.1%–10% action labels.

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